

> WHERE TABLE_SCHEMA = 'momio_sms_and_cycles' ;

CONSTRAINT_NAME	TABLE_NAME	CONSTRAINT_TYPE
PRIMARY	promotions	PRIMARY KEY
PRIMARY	system_logs	PRIMARY KEY
system_logs_ibfk_1	system_logs	FOREIGN KEY
PRIMARY	transaction_types	PRIMARY KEY
PRIMARY	transactions	PRIMARY KEY
transactions_ibfk_1	transactions	FOREIGN KEY
transactions_ibfk_2	transactions	FOREIGN KEY
transactions_ibfk_3	transactions	FOREIGN KEY
transactions_chk_1	transactions	CHECK
PRIMARY	user_promos	PRIMARY KEY
user_promos_ibfk_1	user_promos	FOREIGN KEY
user_promos_ibfk_2	user_promos	FOREIGN KEY
phone	users	UNIQUE
PRIMARY	users	PRIMARY KEY

```
+-----+
| users | CREATE TABLE `users` (
  `user_id` int NOT NULL AUTO_INCREMENT COMMENT 'Unique user identifier',
  `name` varchar(100) DEFAULT NULL COMMENT 'User full name',
  `phone` varchar(20) DEFAULT NULL COMMENT 'User phone number',
  `balance` decimal(15,2) DEFAULT '0.00' COMMENT 'Current account balance',
  PRIMARY KEY (`user_id`),
  UNIQUE KEY `phone` (`phone`)
) ENGINE=InnoDB AUTO_INCREMENT=10 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

```
1 row in set (0.002 sec)
```

```
mysql> █
```

```

+-----+
| transactions | CREATE TABLE `transactions` (
  `transaction_id` varchar(50) NOT NULL COMMENT 'Unique transaction ID from SMS',
  `sender_id` int DEFAULT NULL COMMENT 'User who sent money',
  `receiver_id` int DEFAULT NULL COMMENT 'User who received money',
  `type_id` int NOT NULL COMMENT 'Type of transaction',
  `amount` decimal(15,2) NOT NULL COMMENT 'Transaction amount in RWF',
  `transaction_date` datetime NOT NULL COMMENT 'Date and time of transaction',
  `status` varchar(20) DEFAULT NULL COMMENT 'Transaction status',
  PRIMARY KEY (`transaction_id`),
  KEY `idx_transaction_date` (`transaction_date`),
  KEY `idx_type_id` (`type_id`),
  KEY `idx_sender_id` (`sender_id`),
  KEY `idx_receiver_id` (`receiver_id`),
  CONSTRAINT `transactions_ibfk_1` FOREIGN KEY (`sender_id`) REFERENCES `users` (`user_id`),
  CONSTRAINT `transactions_ibfk_2` FOREIGN KEY (`receiver_id`) REFERENCES `users` (`user_id`),
  CONSTRAINT `transactions_ibfk_3` FOREIGN KEY (`type_id`) REFERENCES `transaction_types` (`type_id`),
  CONSTRAINT `transactions_chk_1` CHECK ((`amount` >= 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+

```

type_name	count	total_amount	avg_amount
Withdrawal	1	20000.00	20000.000000
Cash Power	1	4000.00	4000.000000
Received	2	27000.00	13500.000000
Payment	2	1600.00	800.000000
Bank Deposit	1	5000.00	5000.000000

5. `SELECT COUNT(*) FROM ...`

```
mysql> INSERT INTO transactions (transaction_id, type_id, amount, transaction_date, status)
      -> VALUES ('TEST001', 1, -100.00, NOW(), 'Completed');
ERROR 3819 (HY000): Check constraint 'transactions_chk_1' is violated.
mysql> █
```

```
mysql> INSERT INTO transactions (transaction_id, type_id, amount, transaction_date, status)
-> VALUES ('TEST002', 999, 100.00, NOW(), 'Completed');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`momo_sms_analytics`.`transactions`,
CONSTRAINT `transactions_ibfk_3` FOREIGN KEY (`type_id`) REFERENCES `transaction_types` (`type_id`))
```

```
mysql> INSERT INTO users (name, phone, balance)
-> VALUES ('Test', '250791666666', 1000.00);
ERROR 1062 (23000): Duplicate entry '250791666666' for key 'users.phone'
```

> ORDER BY t.transaction_date DESC,

transaction_id	amount	transaction_date	status	sender_name	receiver_name	transaction_type
14103506143	4000.00	2024-05-26 13:31:00	Completed	Linda Green	NULL	Cash Power
14098463509	20000.00	2024-05-26 02:10:27	Completed	Linda Green	Robert Brown	Withdrawal
43668074924	25000.00	2024-05-14 20:57:36	Completed	Samuel Carter	Linda Green	Received
DEP001	5000.00	2024-05-11 18:43:49	Completed	NULL	Linda Green	Bank Deposit
51732411227	600.00	2024-05-10 21:32:32	Completed	Linda Green	Samuel Carter	Payment
73214484437	1000.00	2024-05-10 16:31:39	Completed	Linda Green	Jane Smith	Payment
76662021700	2000.00	2024-05-10 16:30:51	Completed	Jane Smith	Linda Green	Received

name	promo_name	status	redeemed
Jane Smith	Welcome Bonus	Active	1
Samuel Carter	Loyalty Program	Active	0
Robert Brown	Holiday Special	Expired	1
Alex Doe	Referral Bonus	Active	0
Linda Green	BivaMoMotima	Active	0